
A MICROSCOPIC SMALL-UNCERTAINTY PHASE DIAGRAM FOR AN ENTROPIC-LEARNING SOCIETY

Anonymous authors

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ABSTRACT

We implemented a separate, reproducible calculation of the collective phase diagram of the neural-network society model in the asymptotic regime $C \ll 1$, $V \ll 1$. The calculation keeps the microscopic entropic-learning updates of the model and only replaces the field normalizations $\gamma_C = \sqrt{1 + x^\top C x}$ and $\gamma_V = \sqrt{1 + V}$ by one in the update rule. The resulting simulation sweeps the external control parameters (d, f_d) , where d is the discrimination-field strength and f_d is the fraction of discriminating receivers. We measure the three macroscopic order parameters used by the manuscript: opinion-trust correlation $R_{w,\mu}$, trust-class correlation $R_{\mu,c}$, and opinion-class correlation $R_{c,w}$. The primary discrimination signal is $R_{\mu,c}$. The implementation also retains the full microscopic equations for direct full-versus-reduced validation, logs the maximum deviations of γ_C and γ_V from one, and writes resumable checkpoint files during phase-diagram sweeps.

1 MOTIVATION

The object of interest is a collective phase diagram, not a local phase portrait. The local variables (h_w, h_μ) are interaction-dependent microscopic fields. The phase diagram is instead defined over the external parameters (d, f_d) . For each grid point, a complete many-agent society is simulated until a stationary or statistically stationary regime is reached, and order parameters are measured from the resulting microscopic state.

The small-uncertainty limit is delicate. Setting $C = V = 0$ would remove learning, because the microscopic increments scale as $\Delta w = O(C)$ and $\Delta \mu = O(V)$. We therefore initialize

$$C_I(0) = cI_K, \quad V_{J|I}(0) = v,$$

with small but finite c and v , and expose the ratio $r = v/c$ as a parameter. The default exploratory run uses $c = v = 0.05$.

2 MICROSCOPIC MODEL

Each agent I has a perceptron weight vector $\hat{w}_I \in \mathbb{R}^K$ and covariance C_I . Each directed pair $(J|I)$ has distrust $\mu_{J|I}$ and uncertainty $V_{J|I}$. At each interaction, an issue x , emitter e , and receiver $r \neq e$ are sampled. The emitted opinion is

$$\sigma_e = \text{sign}(\hat{w}_e \cdot x).$$

Only the receiver's ideological state and its directed distrust of the emitter are updated.

The full microscopic fields are

$$h_w = \frac{\sigma_e \hat{w}_r \cdot x}{\gamma_C}, \quad h_\mu = \frac{\mu_{e|r}}{\gamma_V},$$

where $\gamma_C = \sqrt{1 + x^\top C_r x}$ and $\gamma_V = \sqrt{1 + V_{e|r}}$. For a discriminatory receiver, the opinion field entering the modulation functions is shifted to $h_w^D = h_w + D_{e|r}$. The code follows the repository convention in which rows of D index receiver class and columns index emitter class. Case 6 is the default, but all six discrimination matrices are exposed.

3 SMALL- C, V REDUCTION

The reduced dynamics uses $\gamma_C = \gamma_V = 1$ in the microscopic update while retaining finite C and V :

$$\begin{aligned}\hat{w}_r(t+1) &= \hat{w}_r(t) + F_w(h_w^D, h_\mu) C_r x \sigma_e, \\ C_r(t+1) &= C_r(t) + F_C(h_w^D, h_\mu) C_r x x^\top C_r, \\ \mu_{e|r}(t+1) &= \mu_{e|r}(t) + F_\mu(h_w^D, h_\mu) V_{e|r}, \\ V_{e|r}(t+1) &= V_{e|r}(t) + F_V(h_w^D, h_\mu) V_{e|r}^2.\end{aligned}$$

No Landau or phenomenological replacement model is introduced.

The evidence and modulation functions are implemented as

$$\begin{aligned}Z(h_w, h_\mu) &= \Phi(h_w) + \Phi(h_\mu) - 2\Phi(h_w)\Phi(h_\mu), \\ F_w &= \frac{[1 - 2\Phi(h_\mu)]g(h_w)}{Z}, \quad F_\mu = \frac{[1 - 2\Phi(h_w)]g(h_\mu)}{Z}, \\ F_C &= -F_w(F_w + h_w), \quad F_V = -F_\mu(F_\mu + h_\mu),\end{aligned}$$

with a small numerical floor on Z to avoid division by zero in the divergent dissonance corners.

4 ORDER PARAMETERS

For each final microscopic state we compute ideological overlap

$$\rho_{IJ} = \frac{\hat{w}_I \cdot \hat{w}_J}{|\hat{w}_I| |\hat{w}_J|}$$

and trust

$$\eta_{J|I} = 1 - 2\Phi(\mu_{J|I})$$

in the small- V measurement convention. The three primary pair observables are

$$\begin{aligned}R_{w,\mu} &= \frac{1}{N(N-1)} \sum_{J < I} (\eta_{I|J} + \eta_{J|I}) \rho_{IJ}, \\ R_{\mu,c} &= \frac{1}{N(N-1)} \sum_{J < I} G_{IJ} (\eta_{I|J} + \eta_{J|I}), \\ R_{c,w} &= \frac{2}{N(N-1)} \sum_{J < I} G_{IJ} \rho_{IJ}.\end{aligned}$$

Here $G_{IJ} = \kappa_I \kappa_J$ by default, matching the signed class convention already documented in the repository. The most direct discrimination observable is $R_{\mu,c}$: positive values indicate in-group trust and out-group distrust, near-zero values indicate weak class correlation, and negative values indicate reverse discrimination.

5 IMPLEMENTATION

All new code lives in `small-cv-phase-diagram/`, leaving the original `nn-based-simulation/` scripts unchanged. The package provides:

- `smallcv/society.py`: batched microscopic full and small- C, V dynamics;
- `smallcv/sweep.py`: sweeps over (d, f_d) with resumable per-batch checkpoints;
- `smallcv/observables.py`: the three correlations and balance observables;
- `scripts/run_phase_diagram.py`: heatmaps and RGB composite;
- `scripts/compare_full_reduced.py`: convergence of reduced equations to full equations;

- `scripts/convergence_cv.py`: convergence as $c, v \rightarrow 0$ at fixed v/c ;
- `scripts/sensitivity_ratio.py`: sensitivity to $r = v/c$.

The sweep cache key now excludes batch size, because batch size is an execution detail rather than a scientific parameter. A run interrupted after any completed batch can be resumed by rerunning the same command.

6 PRELIMINARY RESULT

A coarse completed run is available at $c = v = 0.05$ on a 9×9 grid with one repeat and a shortened microscopic horizon. It produced continuous heatmaps for $R_{w,\mu}$, $R_{\mu,c}$, and $R_{c,w}$, plus an RGB composite where red encodes $R_{\mu,c}$, green encodes $R_{c,w}$, and blue encodes $R_{w,\mu}$:

```
figures/small_cv_heatmaps_c0.05_r1.png
figures/small_cv_phase_composite_c0.05_r1.png
```

This output is a pipeline and qualitative-inspection artifact, not a final scientific-resolution phase diagram.

A larger 21×21 run with two repeats, $c = v = 0.05$, and `steps-at-c1=50` has completed in checkpointed form. It wrote `data/sweep_small_cv_c0.05_r1_21x21_d3df8249a886.npz` and updated the heatmap and composite figures listed above. The same run can be reproduced or resumed with the following command:

```
python3 scripts/run_phase_diagram.py --n-d 21 --n-fd 21 --repeats 2 --c 0.05
--ratio 1 --steps-at-c1 50 --batch-size 32
```

7 VALIDATION STATUS

Unit tests check the evidence and modulation formulas, the sector symmetry $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$, finite small- C, V learning, and single-interaction convergence of the full and reduced equations as c, v decrease. The current test suite passes:

6 passed.

The code also records maximum observed deviations $|\gamma_C - 1|$ and $|\gamma_V - 1|$ during sweeps, so the asymptotic assumption can be audited for each completed run.

For the completed 21×21 exploratory run, $R_{\mu,c}$ spans approximately $[-0.94, 0.94]$, confirming that the trust-class order parameter captures both reverse and direct discrimination. The diagnostic $\max |\gamma_V - 1|$ remains small, about 0.037, but $\max |\gamma_C - 1|$ reaches a large outlier value. This means the current exploratory run is not yet a uniformly controlled small- C calculation; it is a checkpointed implementation and qualitative map that motivates a follow-up run with smaller c , shorter audited windows, or additional covariance diagnostics.

8 LIMITATIONS AND NEXT STEPS

The present implementation prioritizes correctness and separation from the original simulation scripts. The pure Python interaction loop is slow for large grids at small c because the required horizon scales as $1/c$. The next engineering step is to optimize or parallelize the checkpointed sweep, then generate final heatmaps at multiple decreasing (c, v) values and multiple ratios v/c . Only after those convergence and sensitivity studies should categorical phase-region labels be assigned.